

PROFESSIONAL PROGRAM IN

DATA SCIENCE & ARTIFICIAL INTELLIGENCE

from Machine Learning to Agentic AI

Master analytics, machine learning and deep learning — then build with generative AI and design agentic systems that run end to end.



12 Months
Live + Applied



38 Modules
One Clear Path



12 Projects
Real-World Work



GenAI
+ Agentic AI

Professional Program in

Data Science & AI

CareerZeta's Data Science & AI Professional Program is a 12-month, live and applied program that takes you from data analytics through machine learning and deep learning to generative and agentic AI — so you can build, deploy and govern AI systems end to end.



Certificates from NASSCOM FutureSkills Prime and Microsoft

LOSS epoch 40 val_acc 0.978



CONFUSION



FEATURE IMPORTANCE



Why Data Science & AI?



Every Industry Runs on AI

Banking, healthcare, retail, telecom and manufacturing all use ML to predict, personalize and automate.



Analytics, ML & AI in One Path

Move from data analytics through machine learning and deep learning to generative and agentic AI.



From Notebook to Production

Deploy, monitor and improve models in production — the skill that sets hireable data scientists apart.



A Career Ladder, Not Just a Job

Grow from Data Scientist or ML Engineer to Senior Data Scientist, AI Engineer and Head of AI.



Build With GenAI & Agents

Go beyond using AI tools: build LLM apps, RAG systems and AI agents that do real work.



Proof You Can Show

Graduate with 12 projects and a portfolio an employer can open — not just a certificate.

Program at a Glance



12 Months

Live + applied learning



38 Modules

31 core + 7 Agentic AI



9 Phases

Analytics to AI deployment



12 Projects

Real-world, portfolio-ready



6 Case Studies

Across six industries



3 Pillars

Analytics, ML & AI

About Program

The Data Science & AI Professional Program is built for people who want to build with data and AI — analyze it, model it and put intelligent systems into production. Across nine phases you move from Python, SQL and statistics to machine learning, deep learning and generative AI, then through an Agentic AI immersion, MLOps and an end-to-end capstone.



Learning Format
Live + Applied



12 Months
Live Program



Career Services
By CareerZeta



Certification
NASSCOM & Microsoft

Key Highlights

- ✓ 38 modules across 9 phases
- ✓ 12 real-world projects
- ✓ 6 industry case studies
- ✓ Dedicated GenAI & Agentic AI modules
- ✓ Certificates from NASSCOM FutureSkills Prime & Microsoft
- ✓ For working professionals & graduates
- ✓ Live sessions with industry practitioners
- ✓ 1:1 mentoring sessions
- ✓ Resume & LinkedIn profile building
- ✓ Mock interview preparation
- ✓ Hackathons & peer network
- ✓ No-cost EMI options

Program Pedagogy



Live Instructor-led Sessions

Learn from practitioners who build data and AI systems every day.



Hands-on Projects

Build 12 projects across analytics, ML and AI.



Industry Case Studies

Work through six cases across finance, healthcare, telecom and more.



1:1 Mentoring

Regular check-ins with a mentor to guide your transition.



Hackathons

Solve real problems in teams and showcase your skills.



Peer & Alumni Network

Learn with a cohort and stay connected for referrals and support.



Portfolio Review

Mentor feedback on your capstone and projects before you apply.



GenAI & Agent Labs

Build, test and deploy LLM apps and AI agents.

Who Is This **Program For?**

- ✓ Working professionals moving into data science, ML or AI roles
- ✓ Data and business analysts ready to step up to machine learning
- ✓ Software engineers and developers who want to build AI and LLM applications
- ✓ Graduates who want a hands-on start in data science and AI
- ✓ Anyone who wants to build and deploy AI agents end to end

How to **Join**

Four steps take you from application to your first live session.

1

SUBMIT YOUR APPLICATION

Tell us about your background and why you want to move into data science and AI.



2

SCREENING CALL

A short conversation with our admissions team to understand your goals and confirm program fit.



3

ENROLLMENT & ONBOARDING

Secure your seat, get access to pre-work, and meet your cohort before day one.



4

KICK OFF MODULE 01

Join your first live session and start building from week one.



Flexible Payment

No-cost EMI options available



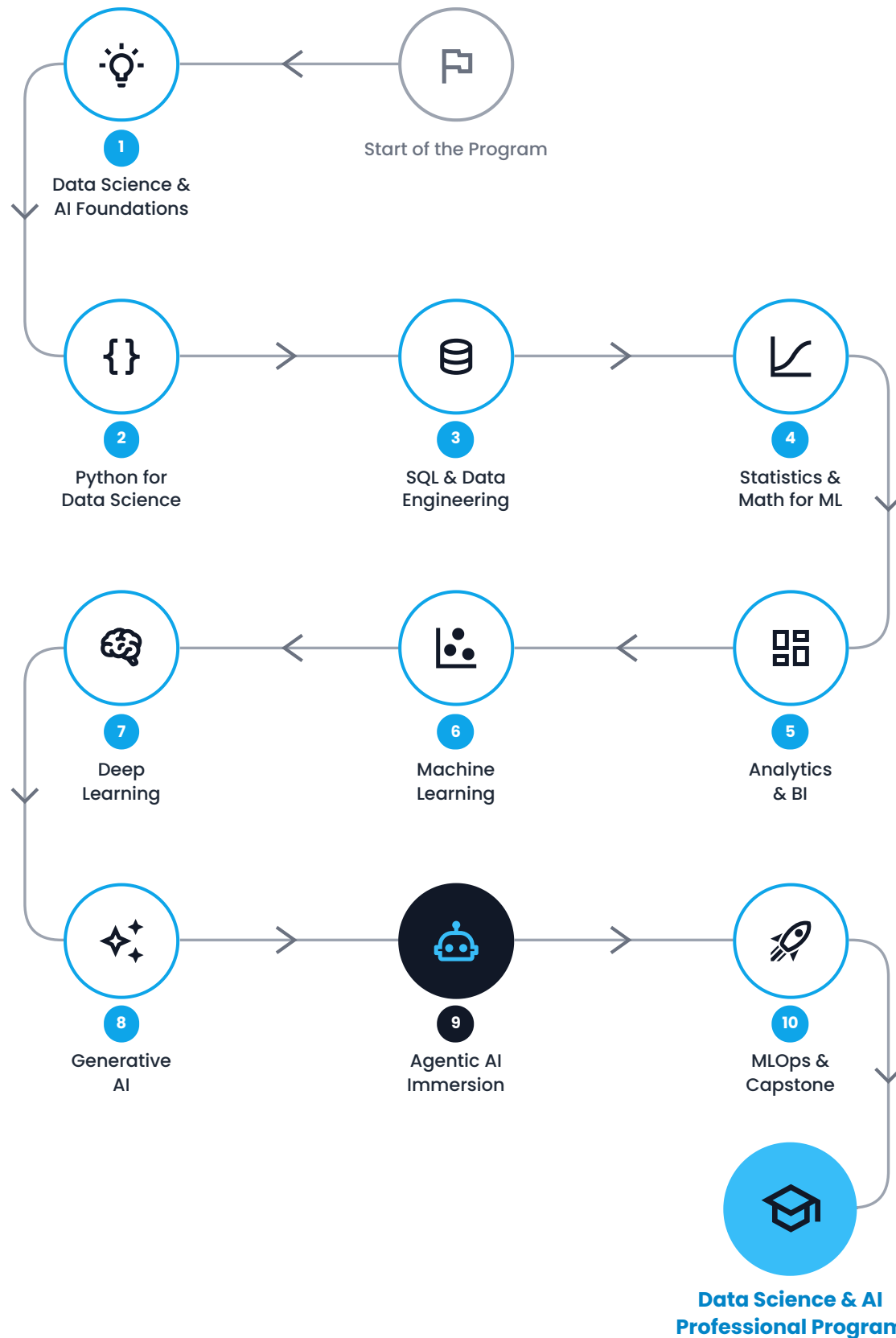
Cohort Size

Kept small for real mentor access

Your Journey

○ Core phases

● Agentic AI



PHASE 1 · DATA SCIENCE & AI FOUNDATIONS

Module 01

Introduction to Data Science & AI

- What Is Data Science? What Is AI?
- Data Analytics vs Data Science vs Machine Learning vs AI
- The Data Science Lifecycle: Problem → Data → Model → Deployment
- Types of ML Problems: Regression, Classification, Clustering, Generation
- Roles in Data & AI Teams
- AI Use-Cases Across Industries
- Responsible AI: Bias, Privacy & Ethics

Module 02

Data Literacy, Data Types & Data Quality

- Structured, Semi-Structured & Unstructured Data
- Data Sources & Collection Methods
- Data Quality Dimensions
- Data Profiling & Validation
- Handling Missing & Inconsistent Data
- Data Documentation & Dictionaries
- Data Privacy & Governance Basics

PHASE 2 · PYTHON FOR DATA SCIENCE

Module 03

Python Programming Foundations

- Python Setup, Jupyter & VS Code
- Variables, Data Types & Operators
- Lists, Dictionaries, Tuples & Sets
- Conditionals, Loops & Comprehensions
- Functions, Modules & Packages
- Error Handling & File Handling
- Object-Oriented Programming Basics
- Writing Clean, Reusable Code

Module 04

Data Wrangling with NumPy & pandas

- NumPy Arrays & Vectorization
- pandas Series & DataFrames
- Indexing, Filtering & Slicing
- Handling Missing Values & Duplicates
- Merging, Joining & Reshaping
- GroupBy & Aggregation
- Working with Dates & Text
- Exploratory Data Analysis Workflow

Module 05

Data Visualization with Python

- Matplotlib Fundamentals
- Statistical Plots with Seaborn
- Interactive Charts with Plotly
- Choosing the Right Chart
- Visualizing Distributions & Relationships
- Data Apps with Streamlit
- Storytelling with Data

PHASE 3 · SQL & DATA ENGINEERING

Module 06

SQL for Data Science

- Relational Databases & Schemas
- SELECT, WHERE, ORDER BY & Aggregations
- GROUP BY & HAVING
- Joins & Subqueries
- CTEs & Window Functions
- Cohort & Funnel Queries
- Query Performance Basics
- Running SQL from Python

Module 07

Data Pipelines & Big Data

- ETL vs ELT
- Data Warehouses & Lakehouses
- Workflow Orchestration with Airflow
- Big Data Processing with PySpark
- Working with APIs & JSON
- Cloud Storage & Compute Basics
- Data Quality in Pipelines

PHASE 4 · STATISTICS & MATHEMATICS FOR ML

Module 08

Descriptive Statistics & Probability

- Mean, Median, Mode & Spread
- Probability Fundamentals
- Conditional Probability & Bayes' Theorem
- Random Variables & Distributions
- Normal Distribution & Z-Scores
- Sampling & the Central Limit Theorem
- Correlation & Covariance

Module 09

Inferential Statistics & Experimentation

- Population vs Sample
- Confidence Intervals
- Hypothesis Testing & p-values
- t-test, Chi-square Test & ANOVA
- A/B Test Design & Analysis
- Statistical Power & Sample Size
- Statistical vs Practical Significance

Module 10

Linear Algebra & Calculus for ML

- Vectors, Matrices & Operations
- Matrix Multiplication & Transformations
- Eigenvalues & Eigenvectors
- Derivatives & Partial Derivatives
- Gradients & the Chain Rule
- Gradient Descent & Optimization
- Loss Functions

PHASE 5 · ANALYTICS & BUSINESS INTELLIGENCE

Module 11

Business Intelligence with Power BI & Tableau

- Data Import & Power Query
- Data Modeling & Relationships
- DAX Fundamentals
- Tableau Calculated Fields & LOD Expressions
- Interactive Reports & Dashboards
- KPI Design & Measurement
- Dashboard Design Best Practices

Module 12

Applied Analytics & Problem Framing

- Turning Business Problems into Data Problems
- Metrics, KPIs & North-Star Metrics
- Customer, Marketing & Product Analytics
- Cohort, Funnel & Retention Analysis
- Root-Cause & Diagnostic Analysis
- Communicating Uncertainty
- Presenting to Stakeholders

PHASE 6 · MACHINE LEARNING

Module 13

Machine Learning Foundations

- What Is Machine Learning?
- Supervised, Unsupervised & Reinforcement Learning
- Train, Validation & Test Splits
- The Bias-Variance Trade-off
- Overfitting & Regularization
- Cross-Validation
- ML Workflows with scikit-learn

Module 14

Feature Engineering & Preprocessing

- Encoding Categorical Variables
- Scaling & Normalization
- Feature Creation & Selection
- Handling Outliers
- Preventing Data Leakage
- Preprocessing Pipelines with scikit-learn

Module 15

Regression Models

- Simple & Multiple Linear Regression
- Polynomial Regression
- Ridge, Lasso & Elastic Net
- Assumptions & Diagnostics
- Evaluation: MAE, RMSE & R²
- Regression for Business Forecasts

Module 16

Classification Models

- Logistic Regression
- k-Nearest Neighbors
- Naive Bayes
- Support Vector Machines
- Decision Trees
- Evaluation: Precision, Recall, F1 & ROC-AUC
- Handling Imbalanced Data

Module 17

Ensembles, Model Tuning & Explainable AI

- Bagging & Random Forest
- Boosting: AdaBoost & Gradient Boosting
- XGBoost, LightGBM & CatBoost
- Stacking & Blending
- Hyperparameter Tuning: Grid, Random & Bayesian Optimization
- Explainability with SHAP & LIME
- Fairness & Bias in Models

Module 18

Unsupervised Learning

- k-means Clustering
- Hierarchical Clustering & DBSCAN
- Principal Component Analysis (PCA)
- t-SNE & UMAP
- Anomaly Detection
- Association Rules & Market Basket Analysis
- Recommender Systems

Module 19

Time Series Forecasting

- Time Series Components
- Stationarity & Differencing
- Moving Averages & Exponential Smoothing
- ARIMA & SARIMA
- Forecasting with Prophet
- ML Models for Time Series
- Forecast Accuracy Metrics

PHASE 7 • DEEP LEARNING

Module 20

Neural Networks & Deep Learning

- Perceptrons & Multi-Layer Networks
- Activation Functions
- Backpropagation & Optimizers
- Dropout, Batch Normalization & Regularization
- Building Models with TensorFlow & Keras
- Building Models with PyTorch
- Training on GPUs

Module 21

Computer Vision

- Image Data & Preprocessing with OpenCV
- Convolutional Neural Networks (CNNs)
- Transfer Learning with Pretrained Models
- Image Classification
- Object Detection with YOLO
- Image Segmentation
- Data Augmentation

Module 22

Natural Language Processing

- Text Preprocessing & Tokenization
- Bag-of-Words & TF-IDF
- Word Embeddings: Word2Vec & GloVe
- RNNs, LSTMs & GRUs
- Sentiment Analysis & Text Classification
- Named Entity Recognition
- NLP with NLTK & spaCy

Module 23

Transformers & Attention

- Attention & Self-Attention
- The Transformer Architecture
- Encoder Models: BERT
- Decoder Models: GPT
- Hugging Face Transformers
- Fine-Tuning Pretrained Models
- Vision Transformers

PHASE 8 · GENERATIVE AI

Module 24

Generative AI & Large Language Models

- How LLMs Work: Tokens, Embeddings & Context
- The LLM Landscape: GPT, Claude, Gemini & Llama
- Open vs Closed Models
- Capabilities, Limits & Hallucinations
- Working with LLM APIs
- Cost, Latency & Token Budgets

Module 25

Prompt Engineering & Structured Outputs

- Prompt Design Principles
- Zero-shot & Few-shot Prompting
- Chain-of-Thought & Reasoning Prompts
- System Prompts & Personas
- Structured Outputs: JSON & Function Calling
- Prompt Evaluation & Iteration

Module 26

Retrieval-Augmented Generation (RAG)

- Embeddings & Semantic Search
- Vector Databases
- Chunking Strategies
- Retrieval, Re-ranking & Hybrid Search
- Building RAG with LangChain & LlamaIndex
- RAG Evaluation

Module 27

Fine-Tuning & LLM Customization

- When to Prompt, Use RAG or Fine-Tune
- Instruction Tuning
- Preparing Training Data
- Parameter-Efficient Fine-Tuning: LoRA & QLoRA
- Evaluating Fine-Tuned Models
- Running Open Models Locally

Module 28

Multimodal & Diffusion Models

- Generative Models: VAEs, GANs & Diffusion
- Text-to-Image with Stable Diffusion
- Vision-Language Models
- Speech-to-Text & Text-to-Speech
- Building Multimodal Apps with Gradio
- Responsible Use of Generative Media

PHASE 9 · MLOPS, DEPLOYMENT & CAPSTONE

Module 29

MLOps & Model Deployment

- The ML Lifecycle & MLOps Principles
- Experiment Tracking with MLflow
- Serving Models with FastAPI
- Containerization with Docker
- CI/CD for Machine Learning
- Monitoring, Drift & Retraining

Module 30

Cloud AI & LLMops

- Cloud ML Platforms: AWS, Azure & GCP
- Infrastructure as Code with Terraform
- Deploying LLM Applications
- LLM Observability & Evaluation
- Cost & Performance Optimization
- Security & Access Control

Module 31

Capstone & AI Portfolio

- End-to-End AI Capstone Project
- Problem Framing & Data Sourcing
- Modeling, Evaluation & Iteration
- Deploying an AI Application or Agent
- Presenting to Stakeholders
- Building a Data Science Portfolio
- Interview-Ready Walkthroughs



Agentic AI Immersion

Seven modules on designing, building and governing AI agents — from single-tool assistants to multi-agent systems that run data science work end to end.

Module A1

Agentic AI Fundamentals

- What Makes a System Agentic
- Agent Architecture: Model, Tools, Memory & Planner
- Agents vs Chatbots vs Workflows
- Single vs Multi-Agent Design
- When to Use — and Not Use — Agents

Module A2

Agent Use-Cases & System Design

- Mapping Data Science Workflows
- AI Opportunity Identification
- Task Decomposition
- Human vs AI Responsibilities
- Feasibility, Risk & ROI

Module A3

Context Engineering for Agents

- Agent Instructions & Operating Rules
- Tool Descriptions & Output Schemas
- Example Trajectories for Agents
- Managing Context Across Long Tasks
- Templates for SQL, Python & ML Agents

Module A4

Tools, APIs & MCP

- Designing Tools for Agents
- Code Execution & Notebook Tools
- Model Context Protocol (MCP) Servers
- Connecting Agents to Databases & BI
- Sandboxing & Permissions

Module A5

Agentic RAG & Knowledge

- RAG as an Agent Tool
- Query Planning & Routing
- Multi-Source & Multi-Hop Retrieval
- Knowledge Graph Basics
- Self-Correcting Retrieval

Module A6

Memory, Planning & Multi-Agent Workflows

- Short- & Long-Term Memory
- Planning & Reflection Loops
- Sequential & Conditional Workflows
- Multi-Agent Orchestration with LangGraph & CrewAI
- Human in the Loop

Module A7

Agent Evaluation, Guardrails & Governance

- Agent Evaluation & Benchmarks
- Trajectory & Tool-Use Checks
- Guardrails & Safety Filters
- Tracing Multi-Step Agent Runs
- Governance, Cost & Human Oversight

Skills to Master

- Python & pandas
- Machine Learning
- NLP & Transformers
- Agentic AI Workflows
- SQL & Data Pipelines
- Deep Learning
- Generative AI & LLMs
- MLOps & Deployment
- Statistics & Probability
- Computer Vision
- RAG & Vector Databases
- Data Storytelling & BI

AI & Data Tools

Gain hands-on experience with industry-leading tools and technologies that power Data Science, AI and Analytics across the globe.



...and many more!

Projects and case studies cover the following industries:



Finance



Healthcare



Retail



E-commerce



Telecom



Manufacturing



Real Estate



Insurance

Beginner

Exploratory Analysis of Retail Sales

Clean and explore two years of retail transactions with pandas. Profile the data, fix quality issues and chart sales by region, category and season to surface the questions the business should act on first.

Beginner

SQL Analytics on an E-commerce Database

Query a multi-table e-commerce database to answer real business questions. Use joins, CTEs and window functions to build customer cohorts, repeat-purchase rates and a revenue funnel a product team can use.

Beginner

House Price Prediction

Build regression models that estimate property prices from location, size and amenities. Compare linear, regularized and tree-based models, and explain which features move the price and by how much.

Intermediate

Credit Default Prediction

Predict loan default on imbalanced lending data. Engineer financial features, handle class imbalance, tune classifiers and choose a decision threshold that balances risk against approval rates.

Intermediate

Customer Churn Prediction with XGBoost

Train gradient-boosted models to flag customers likely to leave. Use SHAP to explain each prediction and turn the drivers of churn into targeted retention actions.

Intermediate

Customer Segmentation with Clustering

Segment customers with k-means and hierarchical clustering on behavioral data. Reduce dimensions with PCA, profile each segment and recommend how marketing should treat each group.

Advanced

Retail Demand Forecasting

Forecast weekly demand across stores and products with ARIMA, Prophet and gradient-boosted models. Measure forecast accuracy and show how better forecasts cut stock-outs and overstock.

Intermediate

Image Classification with Transfer Learning

Classify images with CNNs, then lift accuracy with pretrained models and data augmentation. Compare runs, read confusion matrices and explain where the model still gets it wrong.

Intermediate

Sentiment Analysis of Customer Reviews

Classify reviews by sentiment, moving from TF-IDF baselines to a fine-tuned transformer. Surface the most frequent praise and complaints in a dashboard the business can act on.

Advanced

RAG Assistant for Company Knowledge

Build a retrieval-augmented chatbot over policy documents and FAQs. Chunk and embed content into a vector database, add re-ranking and citations, and measure answer accuracy.

Advanced

Multi-Agent Data Science Crew

Design a team of AI agents that plans an analysis, writes and runs Python, builds charts and drafts the report — with a human approving each step before anything ships.

Advanced

Deploy & Monitor an ML Model

Serve a trained model behind a FastAPI endpoint, containerize it with Docker, track versions in MLflow and set up monitoring that alerts you when data drift starts to hurt performance.

Capstone & **Portfolio**

The program closes with an end-to-end AI capstone: frame a real problem, build the data pipeline, train and evaluate models, and ship an AI application or agent — then shape it all into a portfolio an employer can open.

- ✓ End-to-End AI Capstone Project
- ✓ Problem Framing & Data Sourcing
- ✓ Modeling, Evaluation & Iteration
- ✓ Deploying an AI Application or Agent
- ✓ Presenting to Stakeholders
- ✓ Building a Data Science Portfolio
- ✓ Interview-Ready Walkthroughs

```
ales_2026.csv", parse_dates=["date", "region"]).drop_duplicates()
v(df.date.dt.to_period("M"))["revenue"]
model.predict_proba(X_test)[: , 1]
sts(figsize=(12, 5))
```

```
ment, DATE_TRUNC('month', o.order_value) AS revenue, COUNT(*)
JOIN customers c ON c.customer_id = o.customer_id
date ≥ DATE '2026-01-01'
egment, month
```



Six case studies put the full toolkit to work on the kinds of problems data scientists and AI engineers are hired to solve — each one framed around a real industry question and the decision it has to support.



FINANCE

Card Fraud Detection

Score card transactions for fraud risk. Combine anomaly detection with gradient-boosted classifiers, tune thresholds against the cost of false alarms, and design the monitoring that keeps the model accurate as fraud patterns shift.



HEALTHCARE

Medical Image Triage

Train a deep learning model that flags chest X-rays for priority review. Work with limited labeled data through transfer learning, evaluate sensitivity carefully, and design the workflow so clinicians stay in charge of every decision.



E-COMMERCE

Personalized Recommendations

Grow basket size with recommendations tuned to each shopper. Blend collaborative filtering with product embeddings, test the new engine against the current one in an online experiment and measure the lift in revenue per visit.



TELECOM

Network Anomaly Detection

Monitor network KPIs across thousands of cell sites and catch anomalies before customers notice. Combine time-series baselines with unsupervised anomaly detection, cut false alerts and route each incident to the right engineering team.



MANUFACTURING

Predictive Maintenance

Use sensor time series to predict machine failures before they happen. Engineer rolling features, compare classical and deep learning models, and plan maintenance windows that cut unplanned downtime.



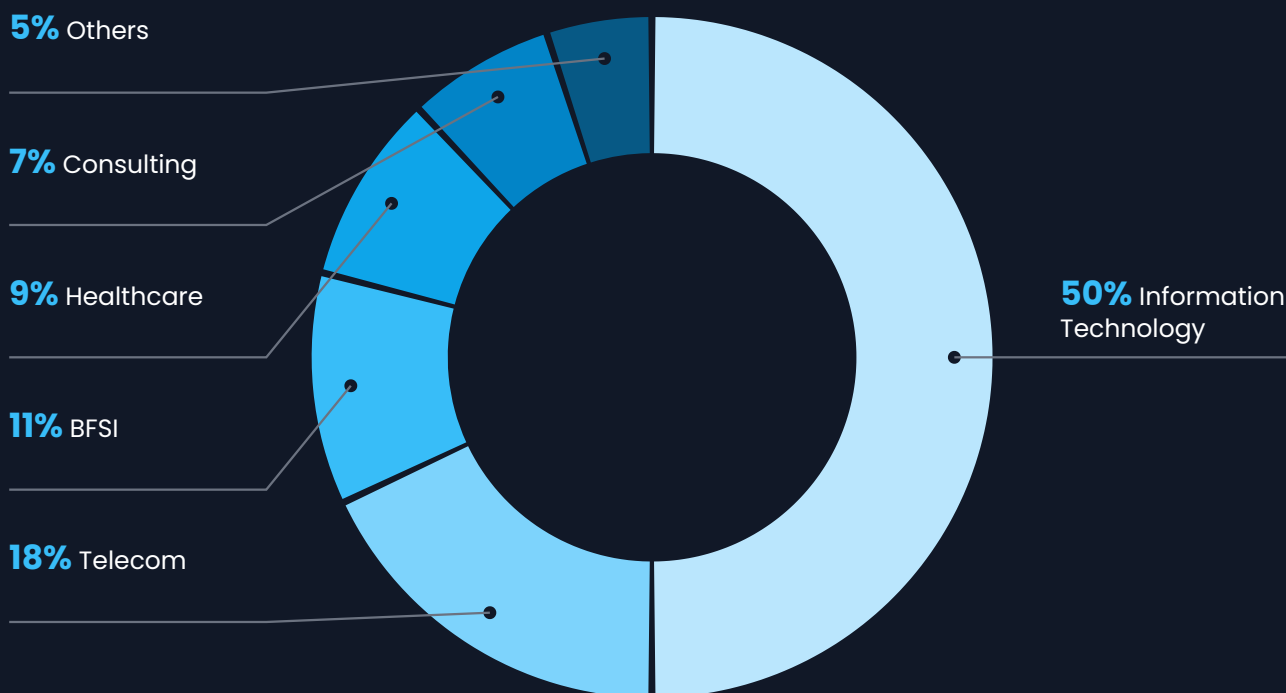
INSURANCE

GenAI Claims & Support Assistant

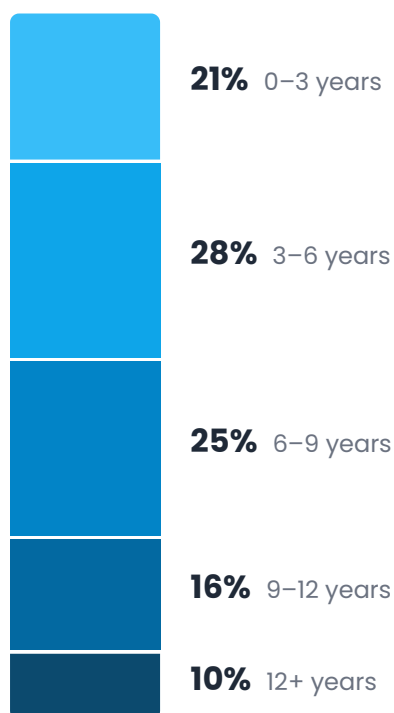
Build an assistant that answers policyholder questions from policy documents and past tickets. Pair RAG with an agent that checks claim status through APIs, add guardrails and hand-off to humans, and measure resolution rates.

Meet the **Batch**

Industries Our Learners Come From



Work **Experience**



Some of their current employers include

amazon

citi

Capgemini

BOSCH

pwc

EY

accenture

Adobe

TATA
TATA CONSULTANCY SERVICES

vmware

Abbott

DELL

CareerZeta Career Services



1:1 Mentoring



Mock Interviews



Portfolio Review



Placements

What We Offer



1:1 Mentoring Sessions

Regular check-ins with a mentor to guide your transition into data science and AI.



Resume & LinkedIn Profile Building

Craft a resume and profile built around the projects you actually shipped in this program.



Hackathons

Work on real-world problems, build solutions in teams, and showcase your skills to industry mentors.



Mock Interview Preparation

Practice with the questions employers actually ask data science and AI candidates.



Career-Oriented Sessions

Live sessions on negotiation, career trajectory planning, and workplace communication.



Portfolio Review

Mentor feedback on your capstone and project portfolio before you start applying.



Alumni & Peer Network

Stay connected to a growing community of CareerZeta learners for referrals and support.



Placement Support

Get access to hiring partners, job opportunities and dedicated support to help you land the right role.

Top Destinations

```
hiring = ["Amazon", "Uber", "Microsoft", "KPMG"]
you.apply(hiring, portfolio=True, mock_prep=True)
support = CareerZeta.placement_support()
# resume + portfolio + practice = interview-ready
print("Next stop: your new ID badge.")
```

Companies across technology, banking, consulting, consumer and fintech that hire for the skills this program builds.



Top Faculty with CareerZeta

```
from careerzeta.faculty import Practitioners
mentors = Practitioners(day_jobs=True)
you.learn_from(mentors, live=True, doubts="all")
# war stories > textbook theory, every time
print("Mentor review: approved. Ship it!")
```

Learn from practitioners who build with data, engineering and AI in their day jobs.



Arnab Das
Data Engineer
leng Group



Anas Qureshi
Technical Lead
NetSkope



Devesh Thapliyal
Technology Lead
Infosys



Ashish Dahiya
Sr. Consultant
Centric Consulting



Rohit Bhatt
Data Management
Engineer
Zurich Insurance



Dileep KH
ML Engineer
University of the West
of England



Rajendra Dhami
Technical Lead
HCL Tech



Divyesh Pandey
Technical Lead
Accenture



Ayush Sharma
Sr. Manager
Angel One

About CareerZeta

CareerZeta is a global edtech platform committed to empowering professionals with industry-relevant, future-ready skills. We deliver practical, career-focused learning experiences designed to help working professionals upskill, advance their careers, and confidently transition into new roles and domains.

Our programs combine expert-led learning, hands-on projects, industry insights, and cutting-edge technology to help professionals stay ahead in today's rapidly evolving digital economy.

Whether you are looking to strengthen your expertise, take the next step in your career, or switch to a new professional profile, CareerZeta provides the skills, guidance, and practical exposure needed to move forward with confidence. We believe professional learning should be practical, accessible, and outcome-focused — helping individuals turn their ambitions into relevant skills, greater confidence, and meaningful career opportunities.

Our Programs



Business Analytics

Turn data into strategic insights and make better business decisions.

+ AGENTIC AI INTEGRATION



Data Analytics

Extract meaningful insights, solve real-world problems, and drive business impact.

+ AGENTIC AI INTEGRATION



Data Science

Build intelligent solutions and apply data-driven thinking to solve real-world challenges.

+ ADVANCE AI INTEGRATION



SAP

Master enterprise solutions and strengthen your expertise in global business operations.

+ AI INTEGRATION



Cybersecurity

Protect systems, manage digital risks, and build a stronger foundation in cybersecurity.

+ AI INTEGRATION



Investment Banking

Develop financial expertise, analyze opportunities, and create value in the world of investment banking.

+ AI INTEGRATION

Ready to **Start?**

Check your eligibility

APPLY NOW

Connect with your admission counselor here:



+91 8851-441177



hello@careerzeta.com



careerzeta.com